

Atmosphere Profiling Drone

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1 Introduction

A sub-200 gram drone payload was developed with full funding support from the Cambridge University Engineering Society. This payload consists of an ESP32-S3 microcontroller, a BNO08x accelerometer, an XHT3x temperature and humidity sensor, an RTC clock module, a microSD card module, and a GPS module (Figure 1). It also includes a 2.4 GHz wireless transmission module for real-time monitoring. The payload was able to probe the atmospheric temperature and humidity profiles directly using the temperature and humidity sensor, as well as the wind velocity profile indirectly through the rotation rate of the payload. This report briefly describes the data retrieval process, which includes synchronising the payload's logging time with the drone's flight log time. The main results presented are nocturnal atmospheric temperature inversion [1] and a characteristic wind profile close to the ground.

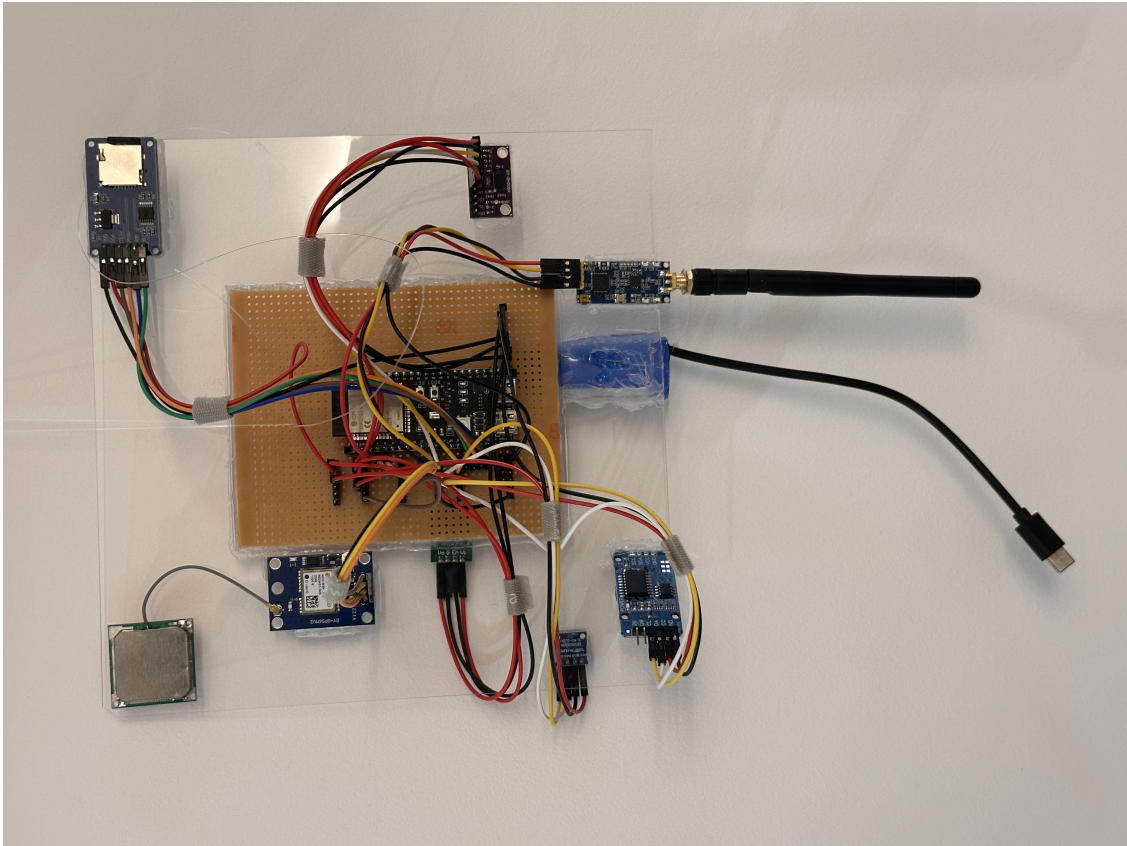


Figure 1: The drone logger. Left to right and top to bottom: microSD card module, accelerometer, 2.4 GHz wireless transmission module, ESP32, battery, GPS module, 3.3V to 5V voltage booster, temp&humidity sensor, RTC clock module.

2 Data Logging and Retrieval

The ESP32 automatically logs accelerometer data, temperature and humidity data, and GPS data onto the microSD card at a theoretical rate of 10 Hz. The accelerometer data include quaternions and gyroscope measurements, with the quaternions being the result of on-chip sensor fusion. The RTC clock module was used to update the system time after each power-on event to generate unique file names. Each data entry also includes a timestamp recorded in microseconds. The logger's boot and shutdown times were recorded during field measurements and used as absolute references for alignment with the DJI drone's flight log obtained via AirData [2].

A Python script was developed to align the data from the logger with the drone data. This was primarily used to provide altitude information for the logger. This was necessary because the logger was not capable of accurately detecting altitude: the accelerometer accumulates errors too rapidly, even after sensor fusion, due to the payload spinning at an average rate of 1.1 to 1.3 turns per second (up to 2.9 turns/s). The logger's GPS also provided unreliable altitude values. In contrast, the drone uses a barometer for altitude determination, which is significantly more accurate.

3 Results

The humidity sensor exhibits a long characteristic response time. As a result, the obtained humidity data show little variation, with transient responses dominating, as shown in Figure 2. A general trend of decreasing relative humidity with increasing altitude may be attributed to changes in temperature. The near-ground relative humidity approaching 100% could be explained by the presence of snow or frost.

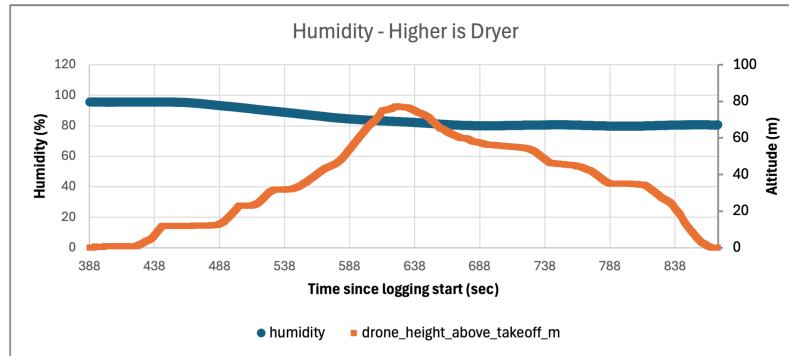


Figure 2: Relative humidity variation with altitude on a clear night.

Further insight was obtained from two measurements of atmospheric temperature profiles. One was conducted on a clear, sub-zero night, while the other was conducted during a cloudy day. The results are shown in Figure 3. It is evident that temperature decreases with altitude during the day and increases with altitude during the night. The latter phenomenon is known as a nocturnal temperature inversion [1]. The recorded temperature gradient (absolute value) was approximately 0.03–0.05 °C/m.

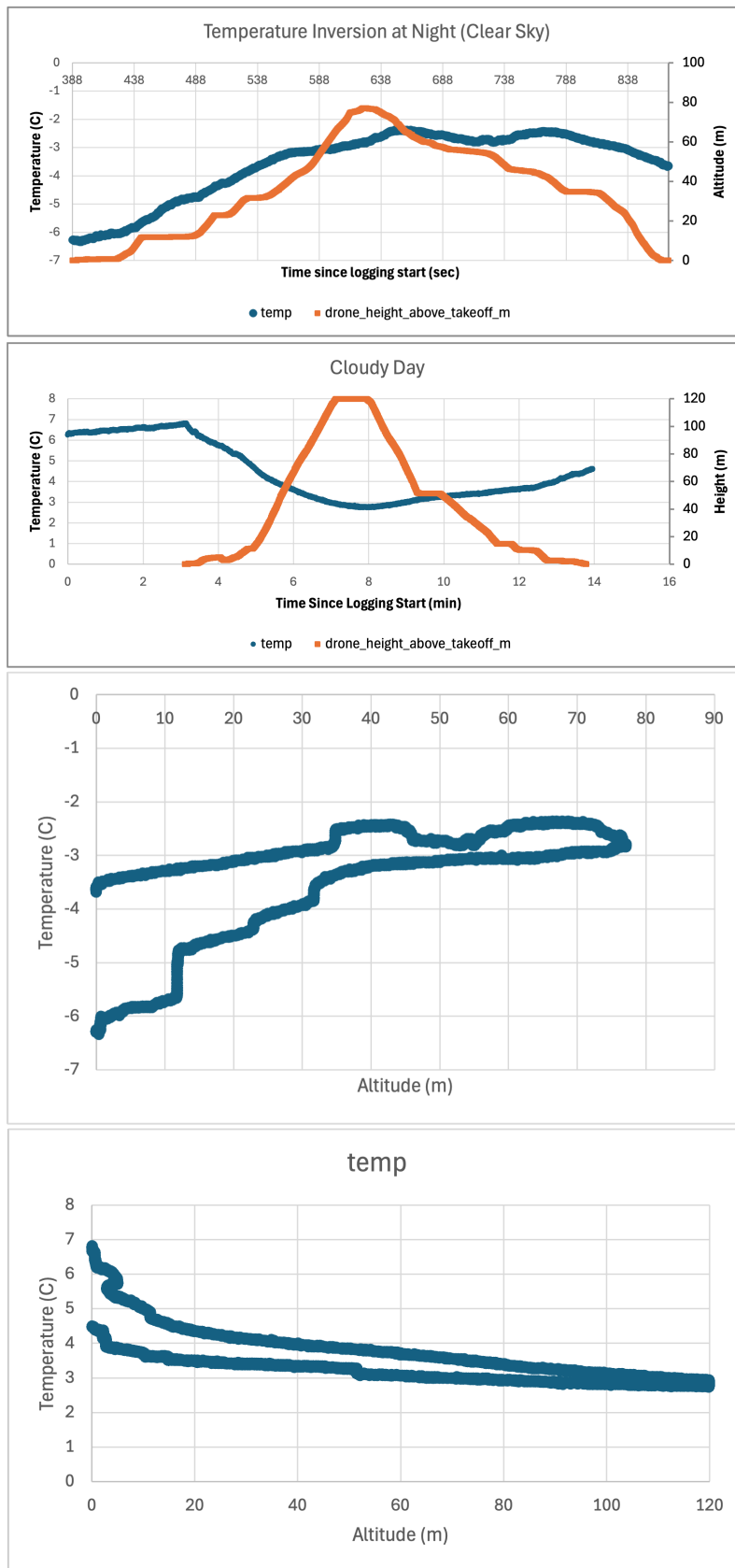


Figure 3: Temperature profiles, from top to bottom: (a) night, (b) day, (c) night, (d) day.

The wind profile in the boundary layer is indicated by the angular velocity of the payload, as shown in Figure 4. The wind velocity increases steadily from the ground up to approximately 10 m in altitude and then remains approximately constant with further increases in altitude. Data logging was terminated either when the drone reached the maximum allowable altitude or when wind velocity increased to the point that the payload and drone became unstable, whichever occurred first.

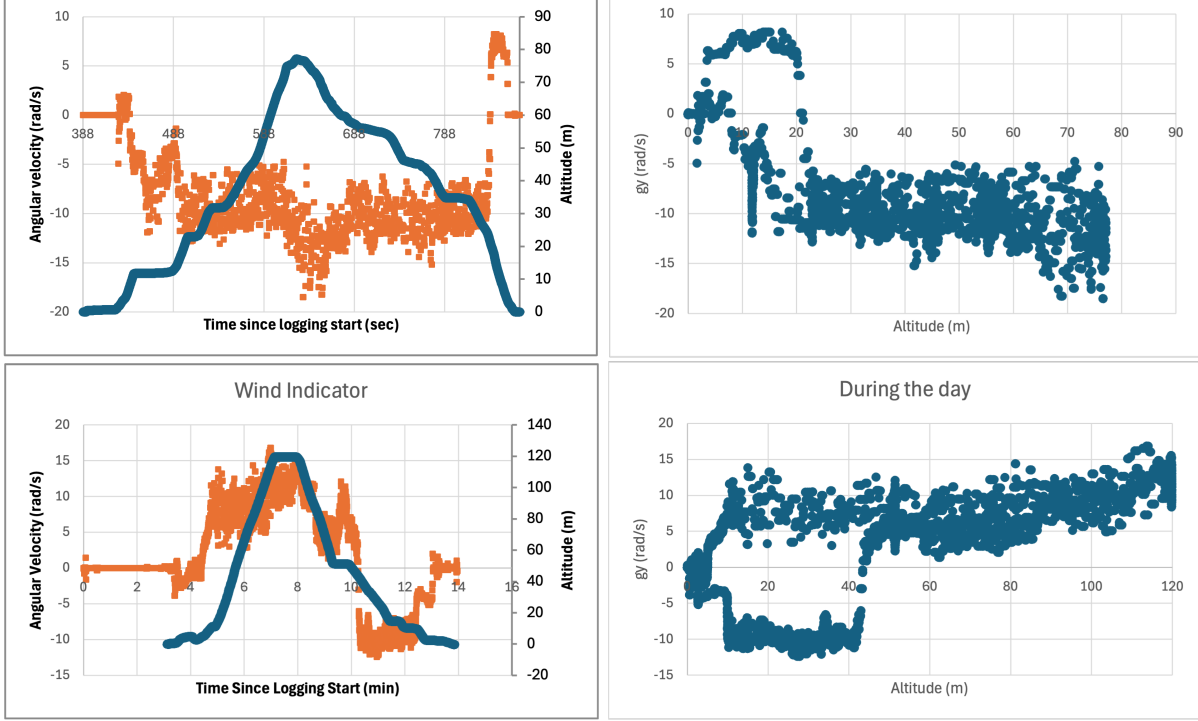


Figure 4: Wind profiles, from left to right and top to bottom: (a) night, (b) night, (c) day, (d) day. The night measurement stopped before the maximum allowable altitude due to stability limitations.

4 Discussions

This project demonstrates the importance of selecting suitable sensors for different measured parameters. Although sensor fusion improves accuracy, it cannot compensate for the intrinsic limitations of individual sensors. For example, altitude determination would be more accurate using a barometer rather than double integration of vertical acceleration from a rapidly spinning accelerometer.

The daytime temperature profile measurement was conducted on a cloudy day to minimise radiative influence from the sun.

The recorded temperature gradients are underestimates due to the presence of transient effects, as evidenced by the differing intercepts in Figure 3(c) and (d). The drone's limited flight time prevented it from hovering long enough for the sensor readings to stabilise fully. Nevertheless, the recorded results are of a similar order of magnitude to measurements obtained in calm atmospheric conditions, where turbulent mixing is minimal [3].

The original proposed method for measuring wind velocity involved using a hot-wire anemometer. This approach was abandoned due to negative reviews regarding the sensor's fragility and reliability. Instead, the wind velocity profile was approximated using the rotational speed of the payload, which increased with wind velocity. Although this method is crude, it still provides qualitative insight into the boundary-layer wind profile. It is less accurate than ultrasonic anemometers and does not provide high-frequency data. The measurements were also affected by torsional energy stored in the fishing line used to suspend the payload, as evidenced by the reversal of rotation as

the payload descended into regions of lower wind speed and reduced driving force.

The observed wind profile suggests that, under stable atmospheric conditions, significant wind shear occurs near the surface. This observation is consistent with established land-based boundary-layer theory [4].

References

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